SmartMatch: A Multi-Agent Pipeline for Mood Board Generation via Graph-Augmented Retrieval and Multimodal Synthesis

Team 04

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Agenda

1. Problem formulation and current limitations
2. System architecture and methods
3. Visual concept grounding and hybrid retrieval
4. Graph-augmented retrieval and set-level optimization
5. Evaluation and ablation results

Demo — Demo and Conclusion

Cross-Modal Retrieval under Abstract Semantic Constraints

Task

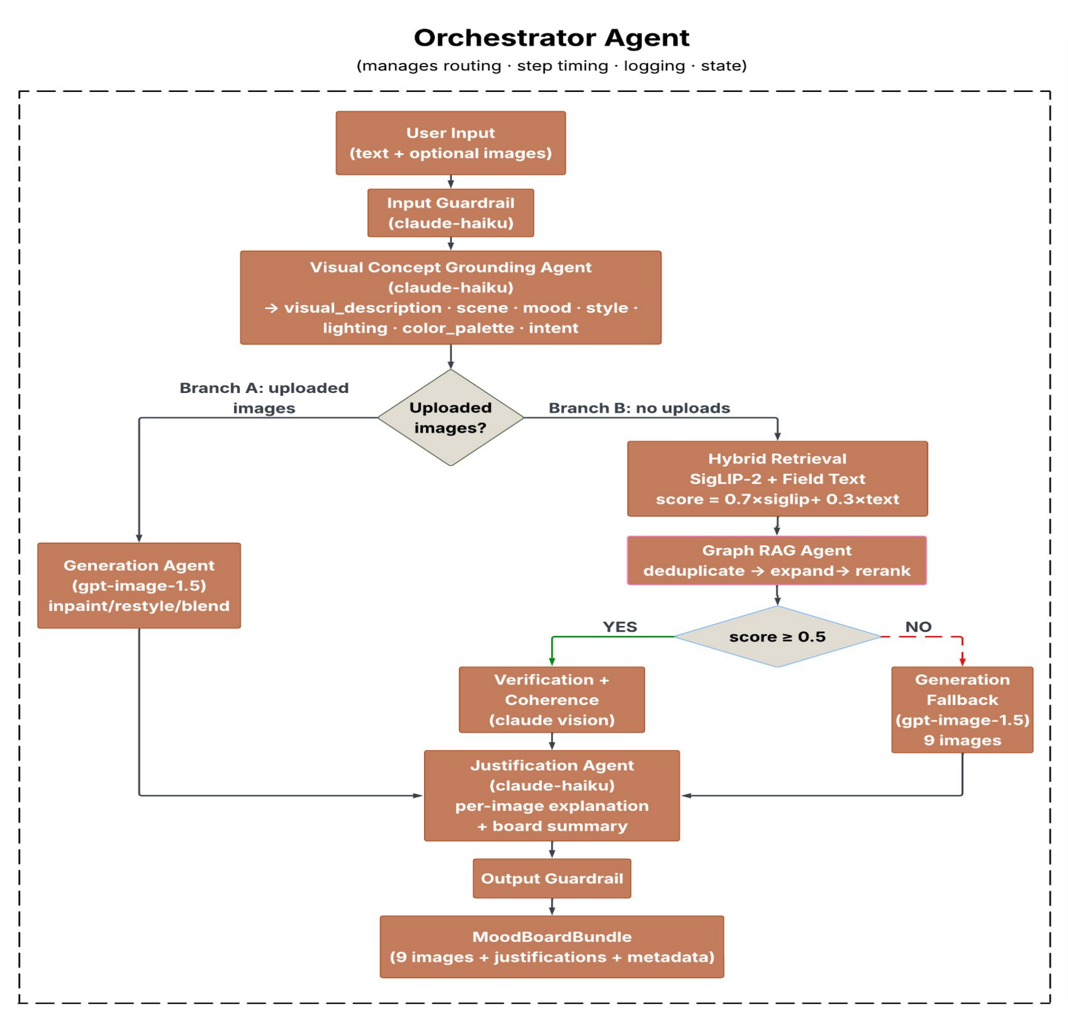
- Map abstract natural language queries → *coherent image sets*

Limitation of existing methods

- Vision-language models optimize **pairwise embedding similarity**
- Assume alignment between textual and visual representations

Core challenge

- Abstract semantics (e.g., emotion, atmosphere) are **not explicitly grounded** in visual embedding space
- ⇒ leads to **systematic cross-modal misalignment**



Pipeline structure

- **Visual grounding** → structured query representation
- **Hybrid retrieval** → candidate image set (multi-signal similarity)
- **Graph RAG** → deduplication, neighbor expansion, reranking
- **Verification + coherence** → set-level selection
- **Generation fallback** → handle low-confidence

Structured Query Representation and Hybrid Retrieval

Visual concept grounding

- LLM transforms input text into structured fields: visual description · mood · color palette · lighting
- Acts as a **multi-field query rewriting mechanism**

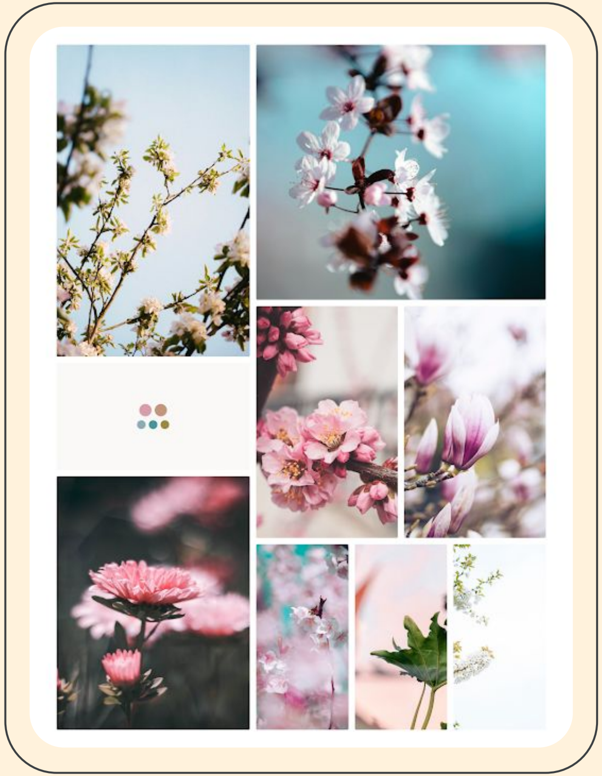
Hybrid retrieval

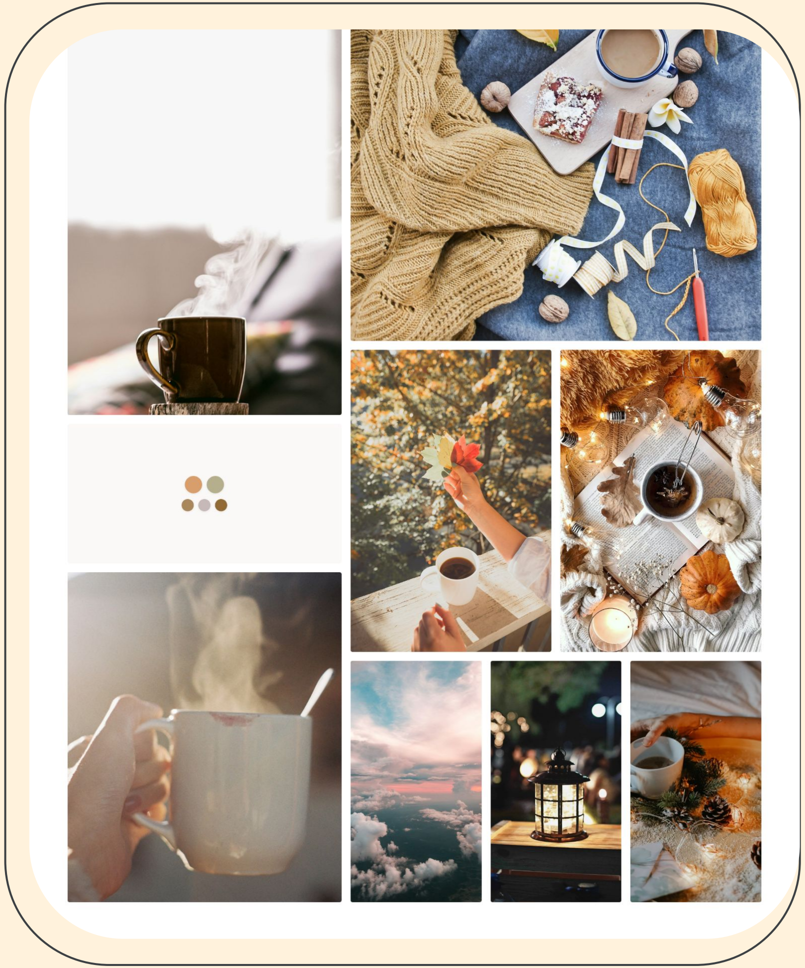
score = 0.3 · s_siglip + 0.7 · s_text

- **SigLIP-2 embeddings** → global image–text similarity
- **Field-level text embeddings** → semantic alignment across attributes

Key property

Text similarity dominates for abstract queries due to weak alignment of visual embeddings in high-dimensional space





Graph-Augmented Retrieval and Set-Level Optimization

Graph construction

- Build weighted graph over image corpus
- Edge weight captures visual and semantic similarity:

w(i,j) = 0.35·sim_mood + 0.20·sim_color + 0.45·sim_visual

Graph operations

- **Deduplication** → remove near-identical candidates
- **Neighbor expansion** → introduce diverse but related images
- **Connectivity-based reranking** → promote well-integrated candidates

Key property

Transforms retrieval into a **set-level optimization problem**, balancing coherence and diversity through graph structure

Set-Level Selection and Adaptive Routing

Coherence-aware selection

- Multimodal verification evaluates alignment between images and query
- Selects final **9-image set** balancing:
  - thematic consistency
  - visual diversity

Set-level scoring

$$s_{final} = (1 - \lambda) \cdot s_{hybrid} + \lambda \cdot s_{graph}$$

- Combines retrieval relevance with graph connectivity
- Promotes images that are **well-integrated in the candidate set**

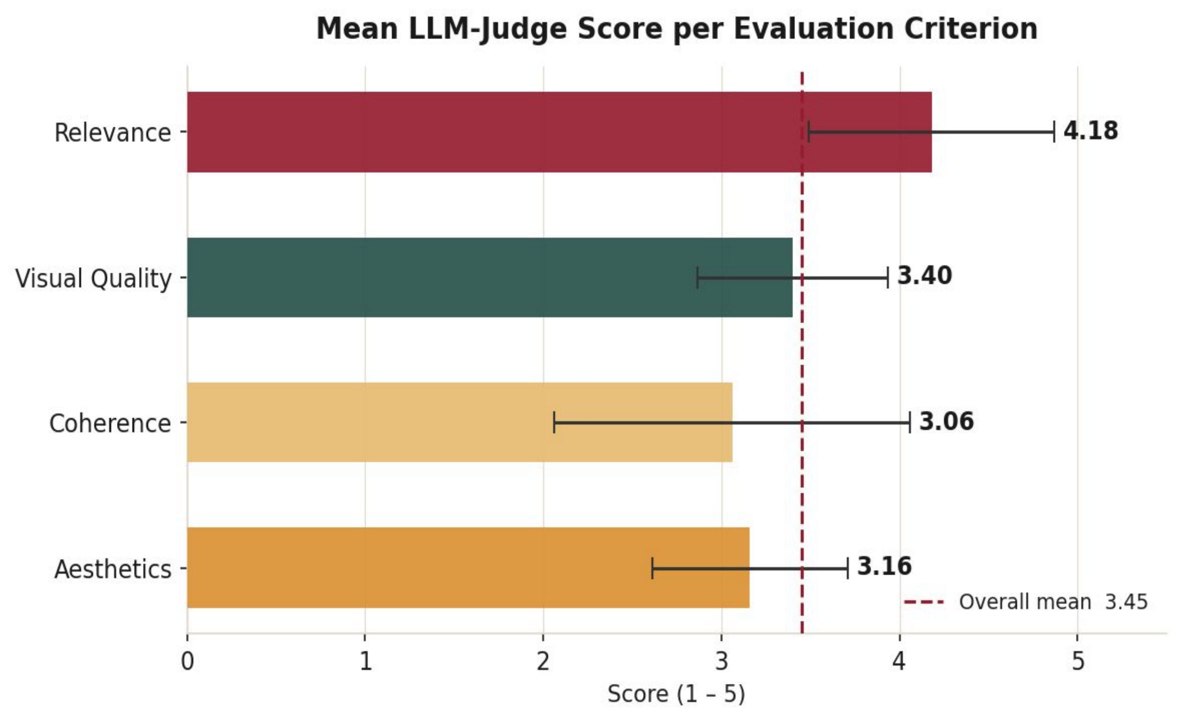
Adaptive routing

- If hybrid score < threshold → **generation fallback**
- Handles queries outside corpus distribution

Key property

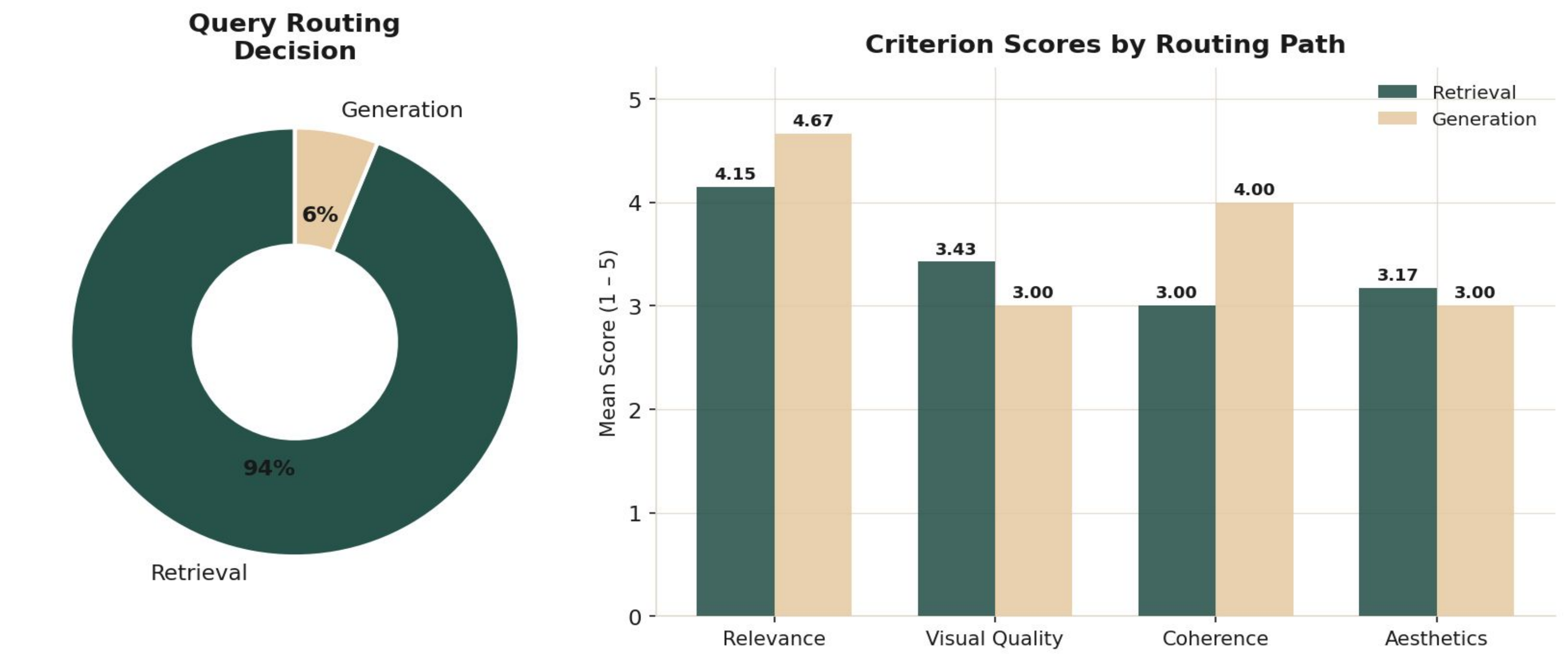
Final output is determined by **joint optimization over candidates**, not independent ranking

Evaluation and Ablation



Mean LLM-judge scores across evaluation criteria (n = 50). Relevance is highest (4.18), while coherence is lowest (3.06), indicating that set-level consistency remains the primary limitation.

Evaluation and Ablation



Query routing is dominated by retrieval (94%), yet generation achieves higher coherence (4.00 vs 3.00), indicating that retrieval-based outputs are limited by corpus homogeneity while generation improves diversity.

Conclusion

- **Grounding** resolves cross-modal misalignment
- **Hybrid retrieval + Graph RAG** enable coherence-aware selection
- **Generation fallback** improves diversity and coherence
- **Limitation:**
  Retrieval yields **visually homogeneous results**

Takeaway

Requires **joint modeling of representation alignment and set structure**, not embedding similarity alone

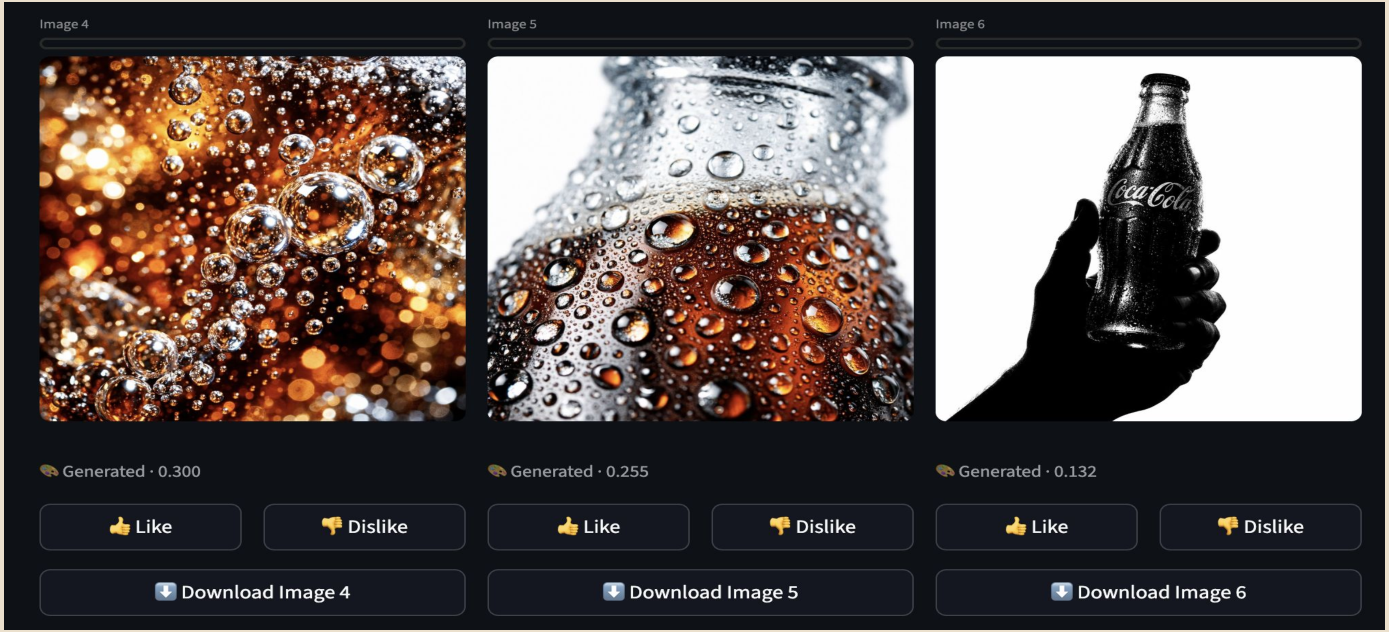

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Image 5
Image 6
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Demo

Q&A

Thank you!